

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

Autonomous Institution Affiliated to Andhra University | Approved by AICTE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Full Stack Web Development Lab

Observation Task: #03

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1. TITLE OF THE EXPERIMENT

Dynamic Student Profile Generator using JavaScript ES6 Classes and DOM Manipulation

2. AIM

To implement a client-side web application that takes student input details, instantiates an ES6 Student class object, and dynamically generates and appends a styled profile card to the DOM upon clicking a button.

3. PROCEDURE / ALGORITHM

Step 1: Create an input form box with fields for Student Name, Roll Number, Department, and CGPA.

Step 2: Define an ES6 Student class with constructor to encapsulate student properties.

Step 3: Attach an onclick event listener to the 'Generate Profile' button.

Step 4: Extract input values from form fields, instantiate a new Student object, and clear previous DOM container content.

Step 5: Dynamically create DOM elements (div.card, h3, div.profile-info) and format using CSS Grid.

Step 6: Append the newly generated card to the DOM container.

4. TEST CASES / VERIFICATION

Test Case	Description / Action	Expected Output	Status
TC-01	Form Input Extraction	Retrieve name, roll, dept, and cgpa from input elements	Passed
TC-02	ES6 Class Instantiation	Instantiate Student object with provided parameters	Passed
TC-03	Dynamic DOM Creation	Create card and profile-info grid elements on the fly	Passed
TC-04	Container Refresh	Clear previous profile before rendering new profile card	Passed

5. LEARNING OUTCOME / RESULT

The Student Profile Generator was successfully designed and tested, demonstrating ES6 OOP concepts, event handling, and dynamic DOM element creation.

6. SOURCE CODE

File: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Student Profile Generator</title>
  <style>
    /* Center everything on the screen */
    body {
      font-family: Arial, sans-serif;
      background: #eef2f3;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
    }

    /* Form Box - Enlarged */
    .form-box {
      background: #ffffff;
      padding: 35px;
      border-radius: 12px;
      box-shadow: 0 6px 15px rgba(0,0,0,0.15);
      text-align: center;
      width: 380px;
    }

    h2 {
      margin-top: 0;
      font-size: 26px;
      color: #333;
    }

    input, button {
      display: block;
      margin: 12px auto;
      padding: 12px;
      width: 90%;
      font-size: 16px;
      border-radius: 6px;
      border: 1px solid #ccc;
      box-sizing: border-box;
    }

    button {
      background: #4CAF50;
      color: white;
      border: none;
      font-weight: bold;
      cursor: pointer;
    }

    /* Generated Profile Card - Enlarged & Colorful */
    .card {
      background: #2c3e50;
      color: white;
      padding: 25px;
      margin-top: 20px;
      border-radius: 10px;
      box-shadow: 0 5px 12px rgba(0,0,0,0.2);
    }

    .card h3 {
      margin-top: 0;
      border-bottom: 2px solid #4CAF50;
      padding-bottom: 8px;
      font-size: 22px;
      color: #4CAF50;
    }
  </style>
</head>
<body>
  <h2>Student Profile Generator</h2>
  <div class="form-box">
    <input type="text" value="Name" />
    <input type="text" value="Email" />
    <input type="text" value="Phone" />
    <input type="text" value="Address" />
    <button type="button" value="Generate Profile" />
  </div>
  <div class="card">
    <h3>Generated Profile</h3>
    <div class="profile">
      <div>Name</div>
      <div>Email</div>
      <div>Phone</div>
      <div>Address</div>
    </div>
  </div>
</body>
</html>
```

```
/* CSS Grid forces perfect left and right alignment */
.profile-info {
  display: grid;
  grid-template-columns: 110px auto;
  row-gap: 10px;
  text-align: left;
  font-size: 17px;
  font-weight: 500;
}
</style>
</head>
<body>

<div class="form-box">
  <h2>Student Profile</h2>
  <input type="text" id="name" placeholder="Name">
  <input type="text" id="roll" placeholder="Roll Number">
  <input type="text" id="dept" placeholder="Department">
  <input type="number" step="0.01" id="cgpa" placeholder="CGPA">
  <button id="btn">Generate Profile</button>
  <div id="container"></div>
</div>

<script>
  // 1. JavaScript Class
  class Student {
    constructor(name, rollNo, department, cgpa) {
      this.name = name;
      this.rollNo = rollNo;
      this.department = department;
      this.cgpa = cgpa;
    }
  }

  // 2. Event Handling & DOM Manipulation
  document.getElementById('btn').onclick = function() {
    // Create Student Object
    const s = new Student(
      document.getElementById('name').value,
      document.getElementById('roll').value,
      document.getElementById('dept').value,
      document.getElementById('cgpa').value
    );

    // Select container and clear previous content
    const container = document.getElementById('container');
    container.innerHTML = '';

    // Dynamic Element Creation
    const card = document.createElement('div');
    card.className = 'card';

    const title = document.createElement('h3');
    title.innerText = 'Student Profile';

    // Grid container for aligned details
    const infoDiv = document.createElement('div');
    infoDiv.className = 'profile-info';

    infoDiv.innerHTML = `
      <span>Name</span><span>: ${s.name}</span>
      <span>Roll No</span><span>: ${s.rollNo}</span>
      <span>Department</span><span>: ${s.department}</span>
      <span>CGPA</span><span>: ${s.cgpa}</span>
    `;

    // Append elements to DOM
    card.appendChild(title);
    card.appendChild(infoDiv);
    container.appendChild(card);
  }
</script>
```

```
};  
</script>  
  
</body>  
</html>
```

7. OUTPUT SCREENSHOTS

Output Screenshot: Screenshot 2026-08-22 194100.png

The screenshot displays a web application interface for creating a student profile. It features a form with four input fields: Name (MANOJ), Roll No (A24126552072), Department (CSM), and CGPA (8). A green 'Generate Profile' button is positioned below the inputs. Below the button, a dark blue box titled 'Student Profile' shows the generated data in a list format.

Student Profile	
Name	: MANOJ
Roll No	: A24126552072
Department	: CSM
CGPA	: 8